MAT 528 HW08

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Due 10/30

Problem 1 (Munkres 30.4)**.** Show that every compact metrizable space X has a countable basis.

Solution. Let d be a metric on X that induces the topology of X. For each $n \in \mathbb{N}$, let $\mathcal{U}_n$ be the collection of open balls in X (with respect to d) with radius $1/n$. This is an open cover of X, so since X is compact it has a finite subcover $\mathcal{B}_n$. Let $\mathcal{B} = \bigcup_{n=1}^{\infty} \mathcal{B}_n$; this is a union of countably many finite sets, so it is countable.

I claim that this collection is a basis for the topology on X. For any $x \in X$ and open $U \subseteq X$ containing x, let $B_r(x)$ be an open ball centered at x with radius r contained in U. Then, pick positive integer m such that $\frac{1}{m} < \frac{r}{2}$. There exists a ball $B_{1/m}(x') \in \mathcal{B}_m$ containing x because $\mathcal{B}_m$ is an open cover of X, and this ball is a subset of $B_r(x)$ because if $y \in B_{1/m}(x')$, then

$$d(x, y) \leq d(x, x') + d(x', y) < \frac{1}{m} + \frac{1}{m} < \frac{r}{2} + \frac{r}{2} = r.$$

Thus given $x \in X$ and $U \subseteq X$ open containing x, there is an element $B \in \mathcal{B}$ such that $x \in B \subseteq U$, which is what it means for $\mathcal{B}$ to be a basis for the topology on X. □

Problem 2 (Munkres 30.9)**.** Let A be a closed subspace of X. Show that if X is Lindelöf, then A is Lindelöf. Show by example that if X has a countable dense subset, A need not have a countable dense subset.

Solution. Let $\mathcal{U}$ be an open cover of A. Each element $U \in \mathcal{U}$ is the intersection of an open set U' with A; let $\mathcal{U}'$ be the collection of these U'. Then, $\mathcal{U}' \cup \{X \setminus A\}$ is an open cover of X, so it has a countable subcover $\mathcal{V}$. If $X \setminus A$ is in this subcover, remove it; the resulting collection of open subsets of X is a countable subset of $\mathcal{U}'$, and since $X \setminus A$ is disjoint from A, $\mathcal{V} \setminus \{X \setminus A\}$ is an open cover of A. Thus $\{V \cap A : V \in \mathcal{V}\}$ is a countable subcover of $\mathcal{U}$, proving that A is Lindelöf.

$\mathbb{R}_\ell^2$ is separable with countable dense subset $\mathbb{Q}^2$, since any basis element $[a, b) \times [c, d)$ contains a point with rational coordinates. On the other hand, the subset $L = \{(a, -a) : a \in \mathbb{R}\}$ is not separable because it is uncountable and inherits the discrete topology from $\mathbb{R}_\ell^2$, as shown in Munkres's Example 30.4. □

Problem 3 (Munkres 31.8)**.** Let G be a compact topological group; let X be a topological space; let α be an action of G on X. If X is Hausdorff, regular, normal, locally compact, or second-countable, then so is X/G.

Solution. We show that the projection map $\pi\colon X \to X/G$ is closed, surjective, and perfect, so that by Munkres's Exercises 31.6 and 31.7, the problem statement holds.

Since $X \to X/G$ is a quotient map, it is surjective. Let $A \subseteq X$ be closed. Then $\pi(A)$ is closed in X/G if and only if $\pi^{-1}(\pi(A))$, which is the union of the orbits of all elements in A, is closed in X. We can write this set equivalently as $GA \coloneqq \{g \cdot x : g \in G, x \in A\}$, which is the image $\alpha(G \times A)$. If $y \notin GA$ then $gy \notin A$ for all $g \in G$. Since $X \setminus A$ is open, for each g there is a neighborhood $U_g \times V_g \subseteq G \times X$ of (g, y) such that $\alpha(U_g \times V_g) \cap A = \varnothing$. The U_g cover G, so we can choose a finite subcover $U_{g_1}, \ldots, U_{g_n}$ because G is compact. Then $V = V_{g_1} \cap \cdots \cap V_{g_n}$ is a neighborhood of y and $\alpha(G \times V)$ does not intersect A, so V does not intersect GA (otherwise, if $z \in V \cap GA$, then $g_0 z \in A$ for some $g_0 \in G$, so $\alpha(G \times \{z\})$ intersects A). This proves that GA is closed, so π is a closed map.

Finally, π is perfect because the orbit of any $x \in X$ is $Gx \coloneqq \{g \times x : g \in G\}$, which is the continuous image of the compact set G, so the preimage of any point $[Gx] \in X/G$, which is $\pi^{-1}([Gx]) = Gx$, is compact. □

Problem 4 (Munkres 32.4)**.** Show that every regular Lindelöf space is normal.

Solution. Let X be regular Lindelöf, and let $A, B \subseteq X$ be disjoint closed subsets of X. Each point x of A has a neighborhood U not intersecting B. Using regularity, choose a neighborhood V of x whose closure lies in U. The collection of all such V (as x varies over A) is an open cover of A, so it has a countable subcover $\{U_n : n \in \mathbb{N}\}$ because closed subsets of Lindelöf spaces are Lindelöf (by Munkres's exercise 30.9). By a similar procedure, choose a countable collection of open sets $\{V_n : n \in \mathbb{N}\}$ covering B such that $\overline{V_n}$ is disjoint from A for each n.

Given n, define

$$U'_n = U_n \setminus \bigcup_{i=1}^{n} \overline{V_i}, \quad V'_n = V_n \setminus \bigcup_{i=1}^{n} \overline{U_i}.$$

Each set U'_n is open because U_n is being intersected with the complement of the closed set $\bigcup_{i=1}^{n} \overline{V_i}$, and U'_n covers A because $\overline{V_i}$ is disjoint from A for all i. Similarly, $\{V'_n : n \in \mathbb{N}\}$ is an open cover of B. Finally, by construction, $\bigcup_{n=1}^{\infty} U'_n$ and $\bigcup_{n=1}^{\infty} V_n$ are disjoint since U'_n is disjoint from $V_j \supseteq V'_j$ for all $j \leq n$ and and V'_n is disjoint from $U_j \supseteq U'_j$ for all $j \leq n$, so we have separated A and B by the open sets $\bigcup_{n=1}^{\infty} U'_n$ and $\bigcup_{n=1}^{\infty} V'_n$. $\square$